

# SatQuery AI — Geospatial Analysis Report

Remote Sensing Intelligence & Multimodal Evidence Verification

|             |                                      |
|-------------|--------------------------------------|
| Job ID:     | a63330b5-ccc7-4210-b378-301f860109aa |
| Project ID: | e64343dc-4fb3-41c3-ac45-61635ae3beaf |
| Task Type:  | GEOGROUND                            |
| Status:     | <b>RUNNING</b>                       |
| Generated:  | 2026-09-26 11:11:52 UTC              |

## 1. Request Specification

Query / Prompt: "locate circles, squares, buildings"

## 2. Factual Findings & Observations

**Finding 1 (geoground\_finding):** Visual grounding completed with OWLv2: detected 2 object regions matching queries [circles, squares, buildings]. Breakdown: 1 circles, 1 squares.

*Measurements:* protocol\_version: 1.0, success: True, job\_id: 75ce51c5-e28b-4ef8-a950-c672dd2b6986, step\_key: geoground\_detection, attempt\_id: 1, output\_assets: {'grounding\_overlay': 'C:\\Users\\HP\\earthquery\\storage\\scratch\\75ce51c5-e28b-4ef8-a950-c672dd2b6986\\grounding\_overlay.png'}, output\_metadata: {'model': 'OWLv2 GeoGround (google/owlv2-base-patch16-ensemble)', 'queries': ['circles', 'squares', 'buildings'], 'total\_detections': 2, 'detections': [{'label': 'circles', 'score': 0.3752, 'box': [63.4, 63.5, 194.9, 194.8]}, {'label': 'squares', 'score': 0.2848, 'box': [179.5, 19.9, 241.1, 61.8]}], 'summary': 'Visual grounding completed with OWLv2: detected 2 object regions matching queries [circles, squares, buildings]. Breakdown: 1 circles, 1 squares.', 'text': 'Visual grounding completed with OWLv2: detected 2 object regions matching queries [circles, squares, buildings]. Breakdown: 1 circles, 1 squares.', 'vram\_allocated\_mb': 1862.86, 'note': 'Open-vocabulary zero-shot visual grounding replacing GeoGround-7B. Operates within 6 GB VRAM limit (~590 MiB allocated).', 'metrics': {'duration\_ms': 4549.68786239624, 'peak\_vram\_mb': 1862.85693359375, 'peak\_ram\_mb': None, 'device\_name': 'cuda', 'cudnn\_enabled': False}, 'error\_code': None, 'error\_message': None

## 3. Operational Boundaries & Limitations

- Observations are generated based on local hardware inference (RTX 5060 Laptop GPU).
- ChangeFormer outputs represent 2D building change masks trained on LEVIR-CD.
- UPerNet semantic segmentation reflects generic ADE20K demo proxy classes unless domain-trained.
- Satellite pixel coordinates are not mapped to geographic bounds unless explicit CRS metadata is verified.
- Analyst confirmation is recommended prior to operational deployment.